

**Ghulam Ishaq Khan Institute of Engineering Sciences and
Technology, Topi**



Semester Project Report
CS-101 Computing and AI

Submitted to

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1. Introduction:

The Library Management System (LMS) is a simple program created using C++ to help manage library books. It allows librarians to keep track of book records and lets students borrow or return books easily. The system uses files to save all data, so the records stay safe even after the program is closed.

2. Purpose:

The primary purpose of the LMS is to:

- Simplify the process of managing library books.
- Provide an easy-to-use interface for both librarians and students.
- Automate book-related operations like adding, borrowing, and returning books.
- Maintain a persistent record of the library inventory.

3. Scope:

3.1.Librarian Functionality:

- Add new books to the system.
- View a catalog of all available books.
- Search for books by their unique IDs.
- Delete books from the library's inventory.
- Save book records to a file for data persistence.

3.2.Student Functionality:

- Borrow books from the library.
- Return borrowed books.
- Search for books by their unique IDs.

- View a catalog of all available books.

3.3.Data Persistence:

- Book records are stored in a file (books.txt) to ensure data consistency across sessions.

3.4.User Interaction:

- A well-structured menu-driven interface for easy navigation.
- Clear messages and prompts for intuitive user interaction.

4. Function details:

Below are the key functions implemented in the program:

4.1. User Roles and Menus

The system distinguishes between two main roles:

- **Librarian:** Access to administrative features for managing the library's inventory.
- **Student:** Access to basic features for borrowing and returning books.

Each role has its dedicated menu for navigating its respective features.

5. Functional Modules

5.1. Main Menu

- **Description:** Serves as the entry point for the application, allowing users to select their role from the following options:
 - Librarian
 - Student
 - Close Application

5.2. Librarian Features

1. Add New Book

- **Description:** Allows the librarian to add new books to the library inventory.
- **Inputs:**
 - Book ID (unique)
 - Book Title
 - Author Name
 - Publication Year
 - Number of Copies
- **Validation:** Checks if the maximum book limit (100) has been reached. If the Book ID already exists and checks if user enters a positive number. Exception Handling to make sure that the Publication year is a positive number as well as the number of copies entered by user are always greater than 0.

2. View All Books

- **Description:** Displays a formatted table of all books in the inventory, including:
 - Book ID
 - Title
 - Author
 - Publication Year
 - Number of Copies

3. Search for a Book by ID

- **Description:** Searches the library's inventory for a specific book using its unique ID.
- **Outputs:** Displays the book's details if found; otherwise, notify the user that the book is not in the inventory.

4. Delete a Book by ID

- **Description:** Removes a book from the library's inventory.
 - Search for the book using its ID.
 - Shift array elements to maintain data consistency.

5. Save Books to File

- **Description:** Writes the current inventory to a file (books.txt) to ensure data persistence.

5.3. Student Features

1. Borrow a Book

- **Description:** Allows students to borrow a book if copies are available.
- **Inputs:**
 - Student Name
 - Registration Number
 - Book ID
- **Validation:**
 - Checks if the book exists and if copies are available.
- **Actions:**

- Decrements the number of available copies upon successful borrowing.

2. Return a Book

- **Description:** Enables students to return a borrowed book.
- **Inputs:**
 - Student Name
 - Registration Number
 - Book ID
- **Actions:**
 - Increments the number of available copies.

3. Search for a Book by ID

- **Description:** Like the librarian's functionality but restricted to viewing details only.

4. View All Books

- **Description:** Displays all books available in the library.

5.4. File Handling

1. Load Books from File

- **Description:** Reads book data from books.txt at the start of the program to populate the inventory.
- **Validation:** Handles cases where the file does not exist.

2. Save Books to File

- **Description:** Saves the current inventory to books.txt for data persistence.

5.5. Utility Functions

1. Clear Screen

- **Purpose:** Clears the console for better user experience.
- **Cross-Platform:** Uses platform-specific commands for Windows and Unix-based systems.

2. Print Line

- **Purpose:** Prints decorative lines for visual clarity.

3. Print Container Border

- **Purpose:** Encases menus and headers in a border for better readability.

4. Center Text

- **Purpose:** Aligns text to the center of the console for consistent formatting.

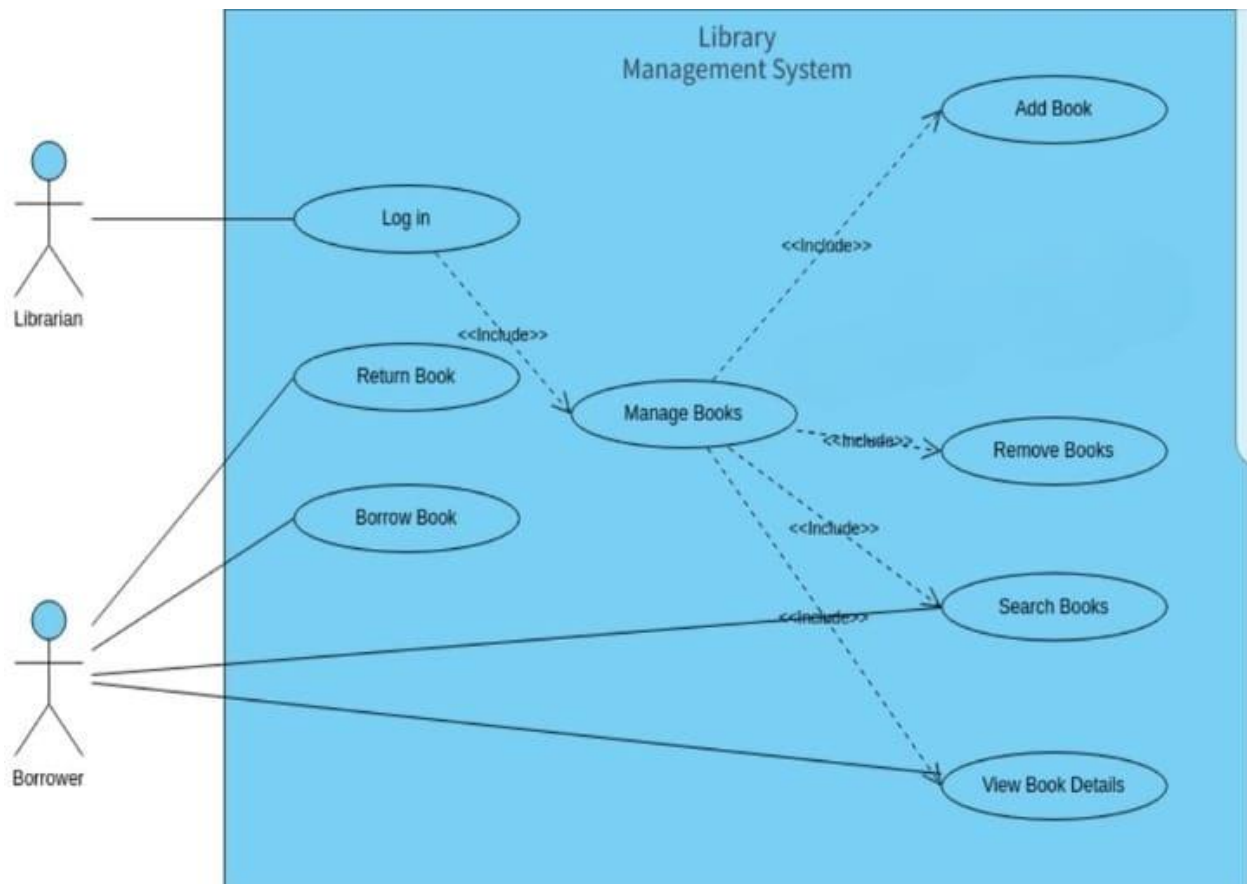
5. Search Book by ID

- **Purpose:** Searches for a book in the inventory by its ID and returns its index (or -1 if not found).

5.6. Constraints

- **Maximum Books:** The system supports up to 100 books due to the array-based implementation.
- **Book ID Uniqueness:** Each book must have a unique ID.+
- **File Name:** The default file for storing and loading book data is books.txt.

6. Use case



7. Screenshots:

Display Main Menu

```
C:\Users\Daniyal\Desktop\lib\library_management.exe
*****
                        LIBRARY MANAGEMENT SYSTEM
*****
                        L M S C++
=====
>> Please Choose Any Option To Login <<

        1. Librarian
        2. Student
        3. Close Application
*****
Enter your choice: _
```

1. If Choice == 1
Librarian Menu

```
C:\Users\Daniyal\Desktop\lib\library_management.exe
*****
                        Librarian Menu
*****

        1. Add New Book
        2. View All Books
        3. Search for a Book by ID
        4. Delete a Book by ID
        5. Save Books to File
        6. Return to Main Menu
*****
Enter your choice: _
```

2. If choice == 2

Student Menu

```
C:\Users\Daniyal\Desktop\lib\library_management.exe
*****
                        Student Menu
*****

        1. Borrow a Book
        2. Return a Book
    3. Search for a Book by ID
        4. View All Books
        5. Return to Main Menu
*****
Enter your choice: _
```

3. If choice == 3

```
C:\Users\Daniyal\Desktop\lib\library_management.exe
*****
                        LIBRARY MANAGEMENT SYSTEM
*****
                        L M S C++
=====
                >> Please Choose Any Option To Login <<

                        1. Librarian
                        2. Student
                    3. Close Application
*****
Enter your choice: 3
Exiting program. Goodbye!

-----
Process exited after 187.5 seconds with return value 0
Press any key to continue . . .
```

Librarian Menu

1. Add book

```
C:\Users\Danial\Desktop\lib\library_management.exe
Enter Book Title: CS101 Proj
Enter Author Name: Danial
Enter Publication Year (positive number): 2024
Enter Number of Copies (positive number): 1

Book added successfully!
*****

Available Books:
-----
  ID              Title              Author        Year   Copies
-----
  1             The Alchemist        Paulo Coelho   1988     10
  2             Moby Dick          Herman Melville 1851      3
  3      To Kill a Mockingbird      Harper Lee     1960      5
  4             CS101 Proj          Danial         2024      1
-----
*****
*****
                          Librarian Menu
*****
                          1. Add New Book
                          2. View All Books
                          3. Search for a Book by ID
                          4. Delete a Book by ID
                          5. Save Books to File
                          6. Return to Main Menu
*****
Enter your choice:
```

1.1. Entering Duplicates of Book ID

```
C:\Users\Danial\Desktop\lib\library_management.exe

Adding a New Book
=====
Enter Book ID (positive number): 4
A book with this ID already exists. Please enter a different Book ID.
Enter Book ID (positive number):
```

2. View All Books

C:\Users\Danial\Desktop\lib\library_management.exe

```
*****
Available Books:
-----
ID           Title           Author        Year    Copies
-----
1           The Alchemist    Paulo Coelho  1988     10
2           Moby Dick       Herman Melville 1851      3
3      To Kill a Mockingbird  Harper Lee    1960      5
4           CS101 Proj       Danial        2024      1
-----
*****
```

3. Search Book By ID

C:\Users\Danial\Desktop\lib\library_management.exe

```
Displaying all books before searching:
*****
Available Books:
-----
ID           Title           Author        Year    Copies
-----
1           The Alchemist    Paulo Coelho  1988     10
2           Moby Dick       Herman Melville 1851      3
3      To Kill a Mockingbird  Harper Lee    1960      5
4           CS101 Proj       Danial        2024      1
-----
*****
Enter Book ID to search: 4
=====
Book Found:
ID           Title           Author        Year    Copies
-----
4           CS101 Proj       Danial        2024      1
=====
```

3.1. Search Book that doesn't exist

```
C:\Users\Daniyal\Desktop\lib\library_management.exe

Displaying all books before searching:
*****

Available Books:
-----
  ID              Title              Author        Year  Copies
-----
   1          The Alchemist        Paulo Coelho  1988    10
   2          Moby Dick          Herman Melville  1851     3
   3    To Kill a Mockingbird      Harper Lee    1960     5
   4          CS101 Proj          Daniyal       2024     1
-----
*****

Enter Book ID to search: 111
=====
Book not found.
=====
*****

                          Librarian Menu
*****

                          1. Add New Book
                          2. View All Books
                          3. Search for a Book by ID
                          4. Delete a Book by ID
                          5. Save Books to File
                          6. Return to Main Menu
*****

Enter your choice:
```

4. Delete a book by ID

```
C:\Users\Daniyal\Desktop\lib\library_management.exe

Displaying all books before deletion:
*****

Available Books:
-----
ID           Title           Author        Year   Copies
-----
1           The Alchemist    Paulo Coelho   1988    10
2           Moby Dick       Herman Melville 1851     3
3           To Kill a Mockingbird Harper Lee     1960     5
4           CS101 Proj      Daniyal        2024     1
-----
*****

Enter Book ID to delete: 3
=====
Book deleted successfully!
=====
*****

Librarian Menu
*****

1. Add New Book
2. View All Books
3. Search for a Book by ID
4. Delete a Book by ID
5. Save Books to File
6. Return to Main Menu
*****

Enter your choice:
```

4.1. Books after deletion of ID: 3

```
C:\Users\Daniyal\Desktop\lib\library_management.exe
*****

Available Books:
-----
ID           Title           Author        Year   Copies
-----
1           The Alchemist    Paulo Coelho   1988    10
2           Moby Dick       Herman Melville 1851     3
4           CS101 Proj      Daniyal        2024     1
-----
*****
*****
```

4.2.If book ID not found

```
C:\Users\Daniyal\Desktop\lib\library_management.exe

Displaying all books before deletion:
*****

Available Books:
-----
ID           Title           Author        Year   Copies
-----
1           The Alchemist    Paulo Coelho  1988    10
2           Moby Dick       Herman Melville 1851     3
4           CS101 Proj      Daniyal      2024     1
-----
*****

Enter Book ID to delete: 444
=====
Book not found.
*****

Librarian Menu
*****

1. Add New Book
2. View All Books
3. Search for a Book by ID
4. Delete a Book by ID
5. Save Books to File
6. Return to Main Menu
*****

Enter your choice: _
```

5. Save books to File (books.txt)

```
C:\Users\Daniyal\Desktop\lib\library_management.exe

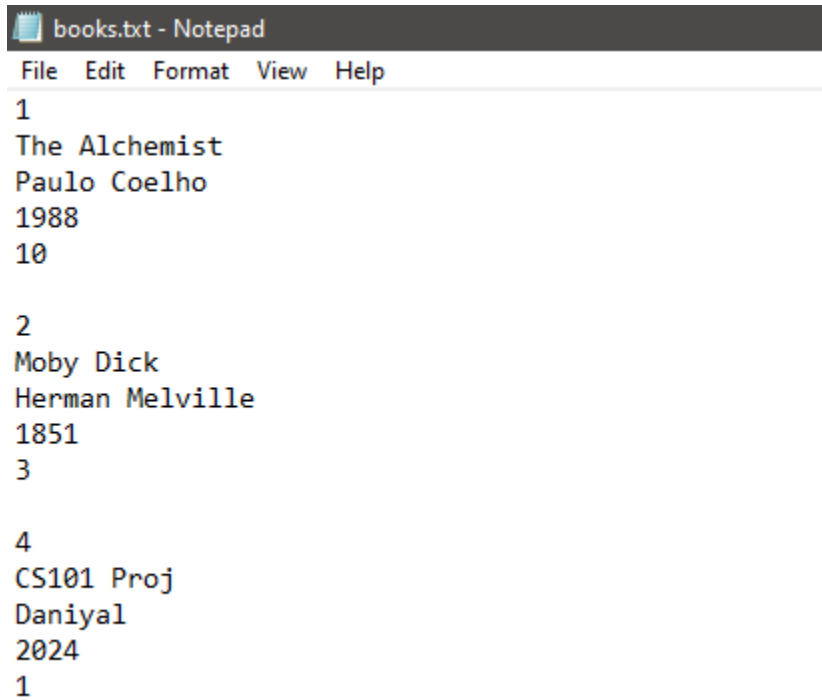
Books saved to file successfully.
*****

Librarian Menu
*****

1. Add New Book
2. View All Books
3. Search for a Book by ID
4. Delete a Book by ID
5. Save Books to File
6. Return to Main Menu
*****

Enter your choice: _
```


Books.txt File after saving Book Id : 4



```
books.txt - Notepad
File Edit Format View Help
1
The Alchemist
Paulo Coelho
1988
10

2
Moby Dick
Herman Melville
1851
3

4
CS101 Proj
Daniyal
2024
1
```

Student Menu

```
C:\Users\Daniyal\Desktop\lib\library_management.exe
*****
                        Student Menu
*****

        1. Borrow a Book
        2. Return a Book
    3. Search for a Book by ID
        4. View All Books
        5. Return to Main Menu
*****
Enter your choice: _
```

1. Borrow a Book

```
C:\Users\Daniyal\Desktop\lib\library_management.exe

Displaying all books before borrowing:
*****

Available Books:
-----
   ID           Title           Author      Year   Copies
-----
   1         The Alchemist      Paulo Coelho 1988     10
   2         Moby Dick         Herman Melville 1851     3
   4         CS101 Proj         Daniyal      2024     1
-----
*****

Enter your name: Daniyal
Enter your registration number: 2024373
Enter Book ID to borrow: 4
=====
Book borrowed successfully by Daniyal (Reg No: 2024373).
=====
*****
                        Student Menu
*****

        1. Borrow a Book
        2. Return a Book
    3. Search for a Book by ID
        4. View All Books
        5. Return to Main Menu
*****
Enter your choice: _
```

1.1.Books Copies in directory after borrowing Book ID: 4

C:\Users\Danial\Desktop\lib\library_management.exe

Available Books:

ID	Title	Author	Year	Copies
1	The Alchemist	Paulo Coelho	1988	10
2	Moby Dick	Herman Melville	1851	3
4	CS101 Proj	Daniyal	2024	0

4.2.Borrowing a book that doesn't exist

C:\Users\Danial\Desktop\lib\library_management.exe

Displaying all books before borrowing:

Available Books:

ID	Title	Author	Year	Copies
1	The Alchemist	Paulo Coelho	1988	10
2	Moby Dick	Herman Melville	1851	3
4	CS101 Proj	Daniyal	2024	1

Enter your name: D
Enter your registration number: 373
Enter Book ID to borrow: 444
=====

Book not found.

2. Return a Book

```
C:\Users\Daniyal\Desktop\lib\library_management.exe

Displaying all books before returning:
*****

Available Books:
-----
ID           Title           Author           Year    Copies
-----
1           The Alchemist    Paulo Coelho     1988     10
2           Moby Dick       Herman Melville  1851      3
4           CS101 Proj      Daniyal          2024      0
-----
*****

Enter your name: Daniyal
Enter your registration number: 2020373
Enter Book ID to return: 4
=====
Book returned successfully by Daniyal (Reg No: 2020373).
=====
*****

                        Student Menu
*****

                        1. Borrow a Book
                        2. Return a Book
                        3. Search for a Book by ID
                        4. View All Books
                        5. Return to Main Menu
*****

Enter your choice:
```

2.1. Book directory after returning

```
C:\Users\Daniyal\Desktop\lib\library_management.exe
```

```
*****
```

Available Books:

ID	Title	Author	Year	Copies
1	The Alchemist	Paulo Coelho	1988	10
2	Moby Dick	Herman Melville	1851	3
4	CS101 Proj	Daniyal	2024	1

```
*****
```

2.2. Returning a book that doesn't exist

```
C:\Users\Daniyal\Desktop\lib\library_management.exe
```

```
Displaying all books before returning:
```

```
*****
```

Available Books:

ID	Title	Author	Year	Copies
1	The Alchemist	Paulo Coelho	1988	10
2	Moby Dick	Herman Melville	1851	3
4	CS101 Proj	Daniyal	2024	1

```
*****
```

Enter your name: D
Enter your registration number: 373
Enter Book ID to return: 444
=====

Book not found.

```
*****
```

3. Search by ID

```
C:\Users\Daniyal\Desktop\lib\library_management.exe

Displaying all books before searching:
*****

Available Books:
-----
  ID              Title              Author          Year   Copies
-----
   1          The Alchemist        Paulo Coelho    1988     10
   2           Moby Dick        Herman Melville  1851      3
   4         CS101 Proj           Daniyal         2024      1
-----
*****

Enter Book ID to search: 4
=====
Book Found:
  ID              Title              Author          Year   Copies
-----
   4         CS101 Proj           Daniyal         2024      1
=====
```

3.1. Searching for a book that doesn't exist

```
C:\Users\Daniyal\Desktop\lib\library_management.exe

Displaying all books before searching:
*****

Available Books:
-----
  ID              Title              Author          Year   Copies
-----
   1          The Alchemist        Paulo Coelho    1988     10
   2           Moby Dick        Herman Melville  1851      3
   4         CS101 Proj           Daniyal         2024      1
-----
*****

Enter Book ID to search: 444
=====
Book not found.
=====
```

Invalid Choice

```
*****
                        Student Menu
*****

        1. Borrow a Book
        2. Return a Book
    3. Search for a Book by ID
        4. View All Books
        5. Return to Main Menu
*****
Enter your choice: 6
Invalid choice! Please try again.
*****
                        Student Menu
*****

        1. Borrow a Book
        2. Return a Book
    3. Search for a Book by ID
        4. View All Books
        5. Return to Main Menu
*****
Enter your choice:
```

8. Code

```
#include <iostream>
#include <fstream>
#include <iomanip>
#include <cstring>
using namespace std;

const int MAX_BOOKS = 100;
const char FILENAME[] = "books.txt";

// Arrays for Books
int bookIds[MAX_BOOKS];
char bookTitles[MAX_BOOKS][50];
char bookAuthors[MAX_BOOKS][50];
int bookYears[MAX_BOOKS];
int bookCopies[MAX_BOOKS];

int bookCount = 0;

// Function Prototypes
void clearScreen();
void displayMainMenu();
void librarianMenu();
void studentMenu();
void loadBooksFromFile();
void saveBooksToFile();
void addBook();
void viewBooks();
```



```

void searchBooks();
void deleteBook();
int searchBookByID(int id);
void incrementCopiesByReference(int &copies);
void borrowBook();
void returnBook();
void printLine(char ch = '=', int length = 75);
void printContainerBorder();
void centerText(const string &text, int width = 75);

// Main Function
int main()
{
    // Load existing books from file
    loadBooksFromFile();

    int choice;
    do
    {
        clearScreen();
        displayMainMenu();
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice)
        {
            case 1:
                clearScreen();
                librarianMenu();

```

```

        break;
    case 2:
        clearScreen();
        studentMenu();
        break;
    case 3:
        cout << "Exiting program. Goodbye!\n";
        break;
    default:
        cout << "Invalid choice! Please try again.\n";
    }
} while (choice != 3);

return 0;
}

// Print a decorative line for = and -
void printLine(char ch, int length)
{
    cout << string(length, ch) << "\n";
}

// Print container border for *
void printContainerBorder()
{
    printLine('*', 75);
}

// Center text within a given width

```

```

void centerText(const string &text, int width)
{
    int padding = (width - text.size()) / 2;

    cout << string(padding, ' ') << text << "\n";
}

// Clear screen
void clearScreen()
{
    system("CLS");
}

// Display the main menu
void displayMainMenu()
{
    printContainerBorder();
    centerText("LIBRARY MANAGEMENT SYSTEM");
    printContainerBorder();

    centerText("L M S C++");
    printLine();

    centerText(">> Please Choose Any Option To Login <<");
    cout << "\n";

    centerText("1. Librarian");
    centerText("2. Student");
}

```

```

        centerText("3. Close Application");

        printContainerBorder();
    }

// Librarian menu
void librarianMenu()
{
    int choice;
    do
    {
        printContainerBorder();
        centerText("Librarian Menu");
        printContainerBorder();

        cout << "\n";

        centerText("1. Add New Book");
        centerText("2. View All Books");
        centerText("3. Search for a Book by ID");
        centerText("4. Delete a Book by ID");
        centerText("5. Save Books to File");
        centerText("6. Return to Main Menu");

        printContainerBorder();

        cout << "Enter your choice: ";
        cin >> choice;
    }
}

```

```
switch (choice)
{
    case 1:
        clearScreen();
        addBook();
        break;

    case 2:
        clearScreen();
        viewBooks();
        break;

    case 3:
        clearScreen();
        searchBooks();
        break;

    case 4:
        clearScreen();
        deleteBook();
        break;

    case 5:
        clearScreen();
        saveBooksToFile();
        break;

    case 6:
        cout << "Returning to Main Menu...\n";
```

```

        break;

        default:
            cout << "Invalid choice! Please try again.\n";
    }
} while (choice != 6);
}

// Student menu
void studentMenu()
{
    int choice;

    do
    {
        printContainerBorder();
        centerText("Student Menu");
        printContainerBorder();

        cout << "\n";

        centerText("1. Borrow a Book");
        centerText("2. Return a Book");
        centerText("3. Search for a Book by ID");
        centerText("4. View All Books");
        centerText("5. Return to Main Menu");
        printContainerBorder();

        cout << "Enter your choice: ";
    }
}

```

```
cin >> choice;

switch (choice)
{
    case 1:
        clearScreen();
        borrowBook();
        break;

    case 2:
        clearScreen();
        returnBook();
        break;

    case 3:
        clearScreen();
        searchBooks();
        break;

    case 4:
        clearScreen();
        viewBooks();
        break;

    case 5:
        cout << "Returning to Main Menu...\n";
        break;

    default:
        cout << "Invalid choice! Please try again.\n";
```

```

        }
    } while (choice != 5);
}

// Load books from file
void loadBooksFromFile()
{
    ifstream inFile(FILENAME);

    if (!inFile)
    {
        cout << "No existing book data found.\n";
        return;
    }

    bookCount = 0;

    while (inFile >> bookIds[bookCount])
    {
        inFile.ignore();
        inFile.getline(bookTitles[bookCount], 50);
        inFile.getline(bookAuthors[bookCount], 50);
        inFile >> bookYears[bookCount] >> bookCopies[bookCount];
        bookCount++;
    }

    inFile.close();
    cout << "Books loaded from file successfully.\n";
}

```



```

// Save books to file
void saveBooksToFile()
{
    ofstream outFile(FILENAME);

    for (int i = 0; i < bookCount; i++)
    {
        outFile << bookIds[i] << "\n"
                << bookTitles[i] << "\n"
                << bookAuthors[i] << "\n"
                << bookYears[i] << "\n"
                << bookCopies[i] << "\n\n";
    }

    outFile.close();
    cout << "Books saved to file successfully.\n";
}

void addBook()
{
    cout << "\nAdding a New Book\n";
    printLine();

    if (bookCount >= MAX_BOOKS)
    {
        cout << "Library is full. Cannot add more books.\n";
        return;
    }
}

```

```

int newBookId;

//Book ID to ensure no duplicates
while (true)
{
    cout << "Enter Book ID (positive number): ";
    cin >> newBookId;

    // Check if the ID already exists
    bool idExists = false;
    for (int i = 0; i < bookCount; i++)
    {
        if (bookIds[i] == newBookId)
        {
            idExists = true;
            break;
        }
    }

    if (idExists)
    {
        cout << "A book with this ID already exists. Please
enter a different Book ID.\n";
    }
    else if (newBookId > 0)
    {
        bookIds[bookCount] = newBookId;
        break;
    }
}

```

```
        else
        {
            cout << "Invalid input! Book ID must be a positive
number.\n";
        }
    }
    cin.ignore();
```

```
    cout << "Enter Book Title: ";
    cin.getline(bookTitles[bookCount], 50);
```

```
    cout << "Enter Author Name: ";
    cin.getline(bookAuthors[bookCount], 50);
```

```
    while (true)
    {
        cout << "Enter Publication Year (positive number): ";
        cin >> bookYears[bookCount];

        if (bookYears[bookCount] > 0)
        {
            break;
        }
        else
        {
            cout << "Invalid input! Publication Year must be a
positive number. Please try again.\n";
        }
    }
}
```

```

        }
    }

    // Validate Number of Copies
    while (true)
    {
        cout << "Enter Number of Copies (positive number): ";
        cin >> bookCopies[bookCount];

        if (bookCopies[bookCount] > 0)
        {
            break;
        }
        else
        {
            cout << "Invalid input! Number of copies must be greater
than 0. Please try again.\n";
        }
    }

    bookCount++;
    cout << "\nBook added successfully!\n";
    viewBooks();
}

// View all books
void viewBooks()
{
    if (bookCount == 0)
    {

```

```

        cout << "No books available in the library.\n";
        return;
    }

    printContainerBorder();
    cout << "\nAvailable Books:\n";
    printLine('-');

    cout << setw(5) << "ID" << setw(30) << "Title" << setw(20) <<
    "Author"
        << setw(10) << "Year" << setw(10) << "Copies\n";

    printLine('-');

    for (int i = 0; i < bookCount; i++)
    {
        cout << setw(5) << bookIds[i]
            << setw(30) << bookTitles[i]
            << setw(20) << bookAuthors[i]
            << setw(10) << bookYears[i]
            << setw(10) << bookCopies[i] << endl;
    }
    printLine('-');
    printContainerBorder();
}

// Search for a book by ID
void searchBooks()
{
    cout << "\nDisplaying all books before searching:\n";

```

```

viewBooks();

int bookId;
cout << "Enter Book ID to search: ";
cin >> bookId;

int index = searchBookByID(bookId);
printLine();
if (index != -1)
{
    cout << "Book Found:\n"
        << setw(5) << "ID" << setw(30) << "Title" << setw(20)
<< "Author"
        << setw(10) << "Year" << setw(10) << "Copies\n";

    cout << setw(5) << bookIds[index]
        << setw(30) << bookTitles[index]
        << setw(20) << bookAuthors[index]
        << setw(10) << bookYears[index]
        << setw(10) << bookCopies[index] << endl;
}
else
{
    cout << "Book not found.\n";
}
printLine();
}

// Search for a book by ID
int searchBookByID(int id)

```

```

{
    for (int i = 0; i < bookCount; i++)
    {
        if (bookIds[i] == id)
        {
            return i;
        }
    }
    return -1;
}

// Delete a book by ID
void deleteBook()
{
    cout << "\nDisplaying all books before deletion:\n";
    viewBooks();

    int bookId;
    cout << "Enter Book ID to delete: ";
    cin >> bookId;

    int index = searchBookByID(bookId);
    printLine();

    if (index == -1)
    {
        cout << "Book not found.\n";
        return;
    }
}

```

```

        for (int i = index; i < bookCount - 1; i++)
        {
            bookIds[i] = bookIds[i + 1];
            strcpy(bookTitles[i], bookTitles[i + 1]);
            strcpy(bookAuthors[i], bookAuthors[i + 1]);
            bookYears[i] = bookYears[i + 1];
            bookCopies[i] = bookCopies[i + 1];
        }
        bookCount--;
        cout << "Book deleted successfully!\n";
        printLine();
    }

// Borrow a book
void borrowBook()
{
    cout << "\nDisplaying all books before borrowing:\n";
    viewBooks();

    string studentName, regNumber;
    cout << "Enter your name: ";
    cin.ignore();
    getline(cin, studentName);

    cout << "Enter your registration number: ";
    getline(cin, regNumber);

    int bookId;

```



```

    cout << "Enter Book ID to borrow: ";
    cin >> bookId;

    int index = searchBookByID(bookId);
    printLine();

    if (index == -1)
    {
        cout << "Book not found.\n";
        return;
    }

    if (bookCopies[index] > 0)
    {
        bookCopies[index]--;
        cout << "Book borrowed successfully by " << studentName
            << " (Reg No: " << regNumber << ").\n";
    }

    else
    {
        cout << "No copies available for borrowing.\n";
    }

    printLine();
}

// Return a book
void returnBook()

```

```

{
    cout << "\nDisplaying all books before returning:\n";
    viewBooks();

    string studentName, regNumber;
    cout << "Enter your name: ";
    cin.ignore();
    getline(cin, studentName);

    cout << "Enter your registration number: ";
    getline(cin, regNumber);

    int bookId;
    cout << "Enter Book ID to return: ";
    cin >> bookId;

    int index = searchBookByID(bookId);
    printLine();
    if (index == -1) {
        cout << "Book not found.\n";
        return;
    }

    bookCopies[index]++;
    cout << "Book returned successfully by " << studentName
        << " (Reg No: " << regNumber << ").\n";
    printLine();
}

```